

Adverse Event Reporting for AI: Developing the Information Infrastructure Government Needs to Learn and Act

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Why Pre-Deployment Testing Alone Cannot Identify All AI Risks

FOR POLICYMAKERS TRYING TO PROACTIVELY ADDRESS AI RISKS, one of the most persistent—and underappreciated—problems is that the most serious risks of advanced AI systems often don’t emerge until after deployment. While much recent attention has focused on pre-deployment risk assessments—testing, evaluation, and red-teaming—these efforts cannot fully anticipate how models will behave in real-world use. Systems like GPT-4, Claude, and DeepSeek continue to surprise even their developers with unexpected capabilities and behaviors post-release. And the uncertainty surrounding model capabilities and risks only grows as general-purpose models are deployed in complex environments.

Key Takeaways

Adverse event reporting systems enable policymakers, industry, and downstream users to learn about AI risks from real-world use.

These systems don’t necessarily require massive new spending or agencies—they can be developed iteratively, scaled over time, and supported through strategic partnerships.

Reporting allows both regulators and industry to respond proactively by surfacing problems quickly, which promotes a culture of safety.

Reporting means better policymaking by providing policymakers with evidence to fill regulatory gaps *only* where they actually exist.

Pre-deployment testing is important, but it is not enough. If policymakers want to ensure that AI development serves the public interest, they need mechanisms that allow government, industry, and society to learn about the technology as it evolves—and to respond when things go wrong.

This is one lesson from earlier governance failures of new and emerging digital technology. Social media platforms were not required to monitor or report harms systematically, and, as a result, policymakers were largely blind to emerging risks until crises like mental health harms provoked reactive responses. AI models may very well follow the same path unless we build capacity to capture and *responsibly* react to new information.

Right now, most of the information about how these systems perform post-deployment is held by private companies, out of reach of policymakers and the public. Closing that gap requires more than asking companies to “do better” with voluntary commitments—it requires building public infrastructure for learning. One central tool for this is **adverse event reporting**.¹

Adverse event reporting systems are already used to surface harms, detrimental events, errors, or malfunctions in other domains. Applied to AI, these systems would provide a structured way to collect reports of model failures, misuse, or unexpected behavior from developers and downstream users. By enabling iterative, evidence-based policymaking, adverse event reporting can help regulators move from guessing about potential risks to understanding what is happening.^{2, 3}

If policymakers are serious about regulating AI in a way that is effective, adaptive, scalable, and sustainable, building an adverse event reporting system should be a top priority. Without it, government risks flying blind.

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Why Reporting Adverse Events Matters: Lessons from Past Experiences

Governments do not need to reinvent the wheel. From monitoring for vehicle safety hazards, aircraft near-misses, adverse vaccine reactions, medical device malfunctions, product defects, data breaches, and cybersecurity risks, incident reporting systems have helped government, industry, and the public identify and respond to risks before harms become widespread. These systems work because they enable policy to move from guesswork to evidence—and they can be built iteratively, scaled with resources, and supported by institutions that already possess the requisite expertise.

An adverse event reporting system provides a more complete depiction of downstream risk, including unanticipated risks. Instead of requiring regulators to specify risks in advance, regulators can define reportable events in terms of damaging or dangerous model output or unanticipated problems to learn about the full range of post-deployment risks—even those that are unknown. In doing so, the system builds an independent evidence base over time.

And, above all, this has worked in practice:

- **Cyber-incident reporting works at scale.**

The Cybersecurity and Infrastructure Security Agency (CISA) has helped detect and fix over 25 million vulnerabilities through its Continuous Diagnostics and Mitigation (CDM) program. Its vulnerability warning pilot has prevented potentially trillions of dollars in damages by identifying over 3 million vulnerabilities across 7,600 organizations. Under the Cyber Incident Reporting for Critical Infrastructure Act (CIRCIA) of 2022, CISA will soon take this further with mandatory reporting requirements to systematically surface risks in real time, improving the agency's ability to proactively warn businesses about critical vulnerabilities.

- **Medical device surveillance is effective—and affordable.** The Food and Drug Administration (FDA) has begun an active postmarket surveillance system for medical devices, to operate in tandem with both mandatory and voluntary adverse event reporting. By analyzing health care data, the active surveillance system complements event reports to identify potential associations between medical devices and adverse health outcomes. The FDA estimates the active surveillance system will cost roughly \$8 million per year to establish and maintain. For less than the cost of product defect recalls, the FDA's active surveillance system supports national monitoring for health and safety risks.

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Why Reporting Matters for AI: Closing the Information Gap

At present, the companies building general-purpose AI systems have more information about the risks than regulators do—which creates a problem. Companies may acknowledge AI-related risks in financial filings, but those summaries are often inaccurate, incomplete, or outdated. Worse, firms may fail to disclose material risks or engage in AI-washing by making misleading claims about AI uses and capabilities.

- **Coordinated responses require reliable information.** Better information doesn't just help policymakers—it helps the entire AI ecosystem. The National Highway Traffic Safety Administration (NHTSA) already uses this approach to monitor for autonomous vehicle safety risks: Developers report accidents daily and regulators can issue recalls based on that data. The same model can work for AI, creating a shared safety infrastructure that benefits both companies deploying AI and downstream users, while encouraging a proactive, coordinated

approach to risk management and cross-platform vulnerabilities. In this way, an adverse event reporting system can help to create a “culture of safety,” where stakeholders prioritize reporting information in an effort to build safer systems—as the aviation sector has successfully done.

- **Identifying necessary regulation requires an informed understanding of risks.** One of the key unknowns in AI regulation is whether we need new laws or whether existing authorities are sufficient to address AI-specific risks. Adverse event reports help answer that question. Reporting systems help lawmakers tailor future legislation to avoid both gaps and unnecessary burdens. Many agencies are equipped to regulate based on the risks of AI without requiring new regulatory or statutory authorities. For example, laws already make malicious uses of technology illegal, and public health, transportation safety, and consumer safety agencies already possess recall authority for devices or technology with adverse safety consequences. A reporting system can help agencies understand how to apply existing regulatory practices to govern AI.

Building a Scalable Adverse Event Reporting System for AI

If policymakers want to regulate AI in a way that addresses this information asymmetry, the first step is clear: We need a system for reporting AI failures and harms after these systems are deployed. Standing up an adverse event reporting system does not require massive resource commitments. The government has created similar systems before—to monitor for

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medical device malfunctions, cybersecurity incidents, and pharmaceutical adverse reactions.⁴

Adverse event reporting systems work because they provide a streamlined, structured way to learn from real-world events. Instead of trying to predict every possible failure in advance—an exercise that is both impossible and fails to capture the magnitude of the risks—these systems allow governments, developers, and researchers to learn iteratively.

1. Start narrow, but learn fast

The first principle for designing a scalable, effective adverse event reporting system for AI is to start narrow, but learn fast. Trying to capture every possible AI risk would overwhelm both agencies fielding reports and mandated reporters. Instead, focusing mandatory reporting on high-consequence risks—like biosecurity or public infrastructure threats—avoids overwhelming the system or degrading the quality of reports.

Clear reporting criteria help reporters and the government. Specific, narrowly scoped reporting

requirements make it easier for agencies to process reports and avoid wasting investigatory resources on noise. As the system matures, reporting criteria should be reassessed regularly to account for new sources of risk. California's data breach notification law provides an illustration of iterative reporting requirements—as it has been amended several times to capture new sources of risk based on emerging trends. Recent amendments to the law include mandatory reporting for breaches of biometric data, for instance.

2. Use a hybrid system to improve detection

A hybrid reporting system may use mandatory reporting requirements on parts of the AI stack where information gathering is most critical, while allowing voluntary reporting to supplement risk assessments and surface rare risks. The FDA's adverse event reporting system for medical devices is one example of a hybrid approach: It collects reports from mandated reporters (manufacturers, importers, and device user facilities) and voluntary reporters (health care professionals, patients, caregivers, and consumers). Similarly, CISA administers a voluntary reporting system for cyber and ransomware incidents. Updates to this reporting system will follow a hybrid model, accepting both voluntary and mandatory reports through the web-based portal once the CIRCIA final rule takes effect.

Employing incentives like liability protections or safe harbors for companies that report quickly and accurately can also increase compliance without requiring heavy-handed enforcement.

3. Don't build from scratch, build partnerships

Partnerships with universities and independent research centers are a proven way to amplify analytical

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capacity at low cost. When the Morris computer worm hobbled internet-connected computers in 1988, unleashing the world's first significant cyberattack, the government did not build a new agency—it partnered with Carnegie Mellon to establish the Computer Emergency Readiness Team Coordination Center (CERT/CC)—a government/academic partnership on cybersecurity incident response that continues to play a key role in cybersecurity. CERT has been successful in leveraging academic expertise to help the government more effectively, and cost-effectively, gain a better understanding of the cybersecurity threat landscape.

The same approach may work for AI. Academic labs can help review reports, surface patterns, and recommend potential remediations, all of which inform policy responses. If properly leveraged, this model can create a continuous feedback loop between the data and government responses.

4. Use information sharing and data access to support analysis

Collecting reports is only a starting point. The next step—integrating the reports into policy debates or

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regulatory decisions—requires directing the reports to actors with the expertise and capacity to investigate them. Agencies should share information with relevant domain experts across government, just as CISA has developed joint ventures like the Joint Cyber Defense Collaborative (JCDC) and Information Sharing and Analysis Organizations (ISAOs) for emerging cyber threats. Summarized, anonymized findings should be published to inform both industry and the public. Open data helps researchers discover risks regulators may miss and shares the load of risk detection and analysis.

Conclusion

Adverse event reporting should play a central role in addressing emerging risks of AI. Other sectors have already demonstrated the utility of such schemes. It's how we learn about dangerous side effects of drugs. It's how we identify automated vehicle risks. It's how we tackle cyber vulnerabilities. And it's a compelling, adaptable, and flexible means for governments to address AI in a way that is grounded in real evidence, not speculation.

Creating an adverse event reporting system for AI need not mean building a cumbersome regulatory apparatus. It means building systems that help the government *learn*. Clear requirements, hybrid reporting, partnerships to support analysis, and information sharing combine to form an agile, adaptable, and scalable framework with the capacity to evolve.

Policymakers do not need to fund moon shot programs or establish new regulatory agencies. They can build a reporting system that learns, adapts, and encourages proactive risk mitigation. They can look to responsible partners to supplement scarce bureaucratic resources. And they can leverage expertise in government and beyond to support and sustain research.

Endnotes

1. See, e.g., Chris W. Johnson, *Failure in Safety-Critical Systems: A Handbook of Incident and Accident Reporting*. Glasgow, UK: University of Glasgow Press (2003); Kevin Paeth, Daniel Atherton, Nikiforos Pittaras, Heather Frase, and Sean McGregor, “Lessons for Editors of AI Incidents from the AI Incident Database,” ArXiv preprint (2024), <https://arxiv.org/abs/2409.16425>.
2. The National Artificial Intelligence Advisory Committee (NAIAC). Recommendation: Improve Monitoring of Emerging Risks from AI through Adverse Event Reporting. 2023.
3. Rishi Bommasani, Scott R. Singer, Ruth E. Appel, Sarah Cen, A. Feder Cooper, Elena Cryst, Lindsey A. Gilmard, Ian Klaus, Meredith M. Lee, Inioluwa Debora Raji, Anka Reuel, Drew Spence, Alexander Wan, Angelina Wang, Daniel Zhang, Daniel E. Ho, Percy Liang, Dawn Song, Joseph E. Gonzalez, Jonathan Zittrain, Jennifer Tour Chayes, Mariano-Florentino Cuéllar, Li Fei-Fei. “The California Report on Frontier AI Policy.” The Joint California Policy Working Group on AI Frontier Models. June 17, 2025.
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